

	Work Instruction	Doc. Revision	01
	AXI V810 PSP 2 Triggering Cable Checking Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

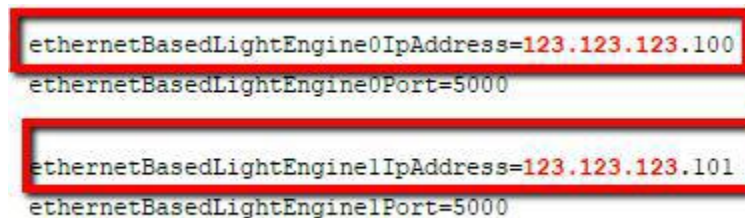
REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	17 Dec 2014	First Release	ONG KEAN SHENG
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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1. Close V810 Software.
2. Go to HW Config (May refer to below table) and look for below Hardware Key.
3. Check & record down the IP address for both PSP 2 projectors. Remarks:

Machine Model	HW Config Path
V810 Strd (Series 1 & Series 2)	C:\Program Files\AXI System\v810\5.0\config\hardware.config
V810 XXL (Series 1& Series 2)	C:\Program Files\AXI System\v810\5.0\xxlconfig\hardware.config
V810 S2EX	C:\Program Files\AXI System\v810\5.0\s2exConfig\hardware.config



```

ethernetBasedLightEngine0IpAddress=123.123.123.100
ethernetBasedLightEngine0Port=5000

ethernetBasedLightEngine1IpAddress=123.123.123.101
ethernetBasedLightEngine1Port=5000

```

Figure 1 PSP 2 projector IP address at HW Config (Reference Picture Only)

Special remarks: Above IP address is just for reference only, end user will required to use the exact IP address from the dedicated HW Config as those IP address might varies from each machine.

IP address end with ".100 means Front Module "; IP address end with ". 101 means Rear Module".

May refer to below table:

Type	IP ADDRESS (Front MODULE)	IP ADDRESS (REAR MODULE)
OLD MACHINE	192.168.0.100	192.168.0.101
NEW MACHINE	123.123.123.100	123.123.123.101

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4. Launch the “Launch StTrgDisplayVC2005.exe “

	Ethernet Bootloader	8/28/2014 12:08 PM	File folder	
	dotNetFx40_Client_x86_x64.exe	10/22/2013 3:17 PM	Application	41,993 KB
	PSP 2 software.7z	1/29/2016 2:20 PM	7Z File	126,914 KB
	StCamSWare_x64.exe	4/4/2014 8:38 AM	Application	11,318 KB
	StTrgDisplayVC2005.exe	2/4/2013 10:25 AM	Application	595 KB
	uEye64_38200_WHQL.exe	4/4/2014 8:37 AM	Application	79,672 KB

Figure 2 StTrgDisplayVC2005 File

5. If error prompt after launching this software where complaining “**Missing. Net Framework**”, please install “**dotNetFx40_Full_x86_x64.exe** “.

If able to open “**StTrgDisplayVC2005.exe** “, please direct go to Step 5.

PSP 2 software			
Name	Date modified	Type	Size
	Ethernet Bootloader	8/28/2014 12:08 PM	File folder
	dotNetFx40_Client_x86_x64.exe	10/22/2013 3:17 PM	Application 41,993 KB
	PSP 2 software.7z	1/29/2016 2:20 PM	7Z File 126,914 KB
	StCamSWare_x64.exe	4/4/2014 8:38 AM	Application 11,318 KB
	StTrgDisplayVC2005.exe	2/4/2013 10:25 AM	Application 595 KB
	uEye64_38200_WHQL.exe	4/4/2014 8:37 AM	Application 79,672 KB

Figure 3 Dot Net Software

6. Camera Select dialog will display. Please select the appropriate camera.

Remarks: Camera 1 – Front Module Camera, Camera 2 –Rear Module Camera

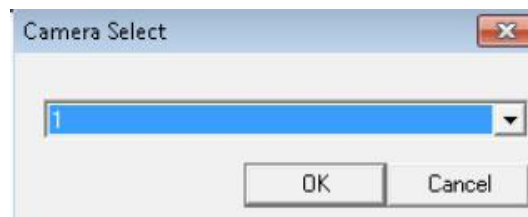


Figure 4 Camera Selection

7. A dialog will be displayed. Go to Mode->Setting

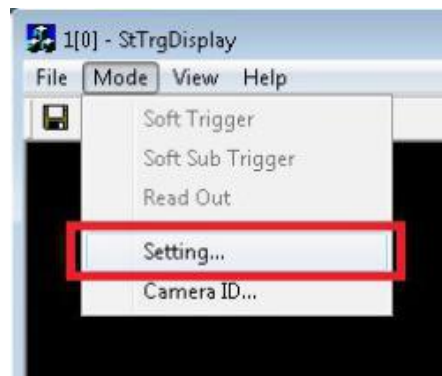
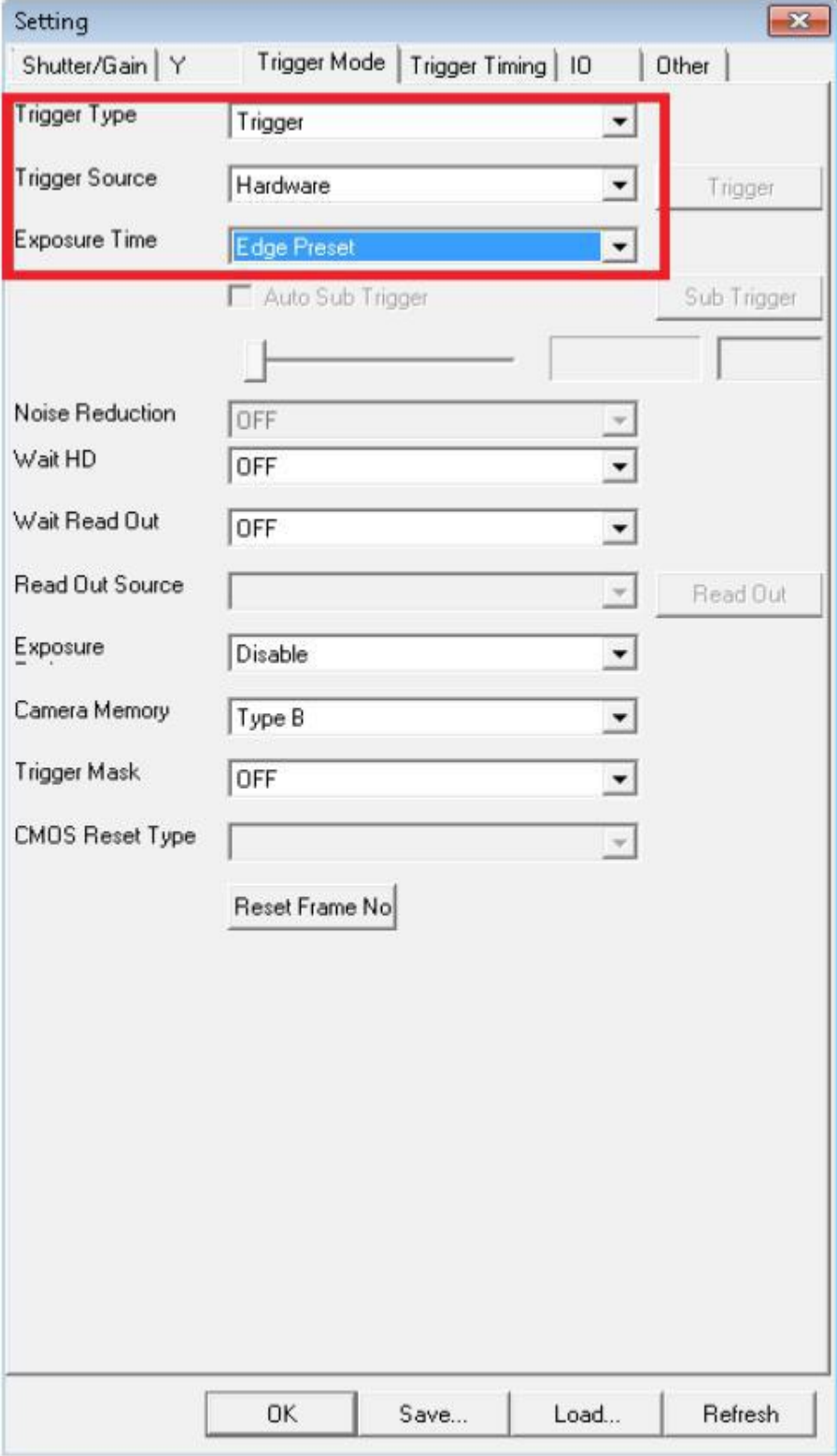


Figure 5 StTrg Display GUI

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8. Another New Dialog box will be appearing. Under Trigger tab, make sure the settings are set as below Figure 6.



The screenshot shows the 'Setting' dialog box with the 'Trigger' tab selected. The following settings are visible:

- Trigger Type:** Trigger (dropdown menu)
- Trigger Source:** Hardware (dropdown menu)
- Exposure Time:** Edge Preset (dropdown menu)
- Auto Sub Trigger:** ☐ (checkbox)
- Noise Reduction:** OFF (dropdown menu)
- Wait HD:** OFF (dropdown menu)
- Wait Read Out:** OFF (dropdown menu)
- Read Out Source:** (dropdown menu)
- Exposure:** Disable (dropdown menu)
- Camera Memory:** Type B (dropdown menu)
- Trigger Mask:** OFF (dropdown menu)
- CMOS Reset Type:** (dropdown menu)
- Reset Frame No:** (text input field)

Buttons on the right side of the dialog include 'Trigger', 'Sub Trigger', and 'Read Out'. At the bottom, there are buttons for 'OK', 'Save...', 'Load...', and 'Refresh'.

Figure 6 Setting

8. Next go to IO tab at row I00. Choose “Trigger Input” as below Figure 7.

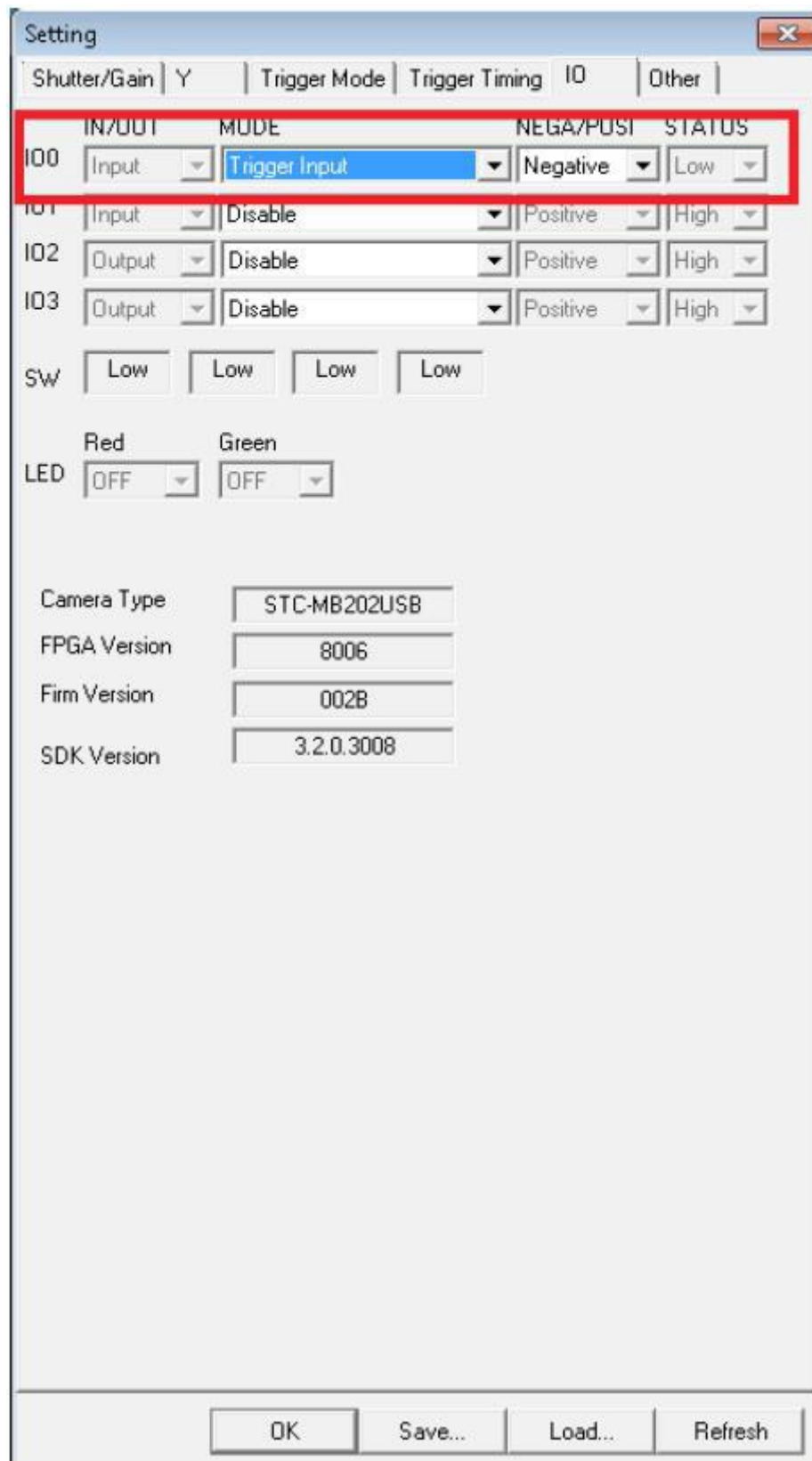


Figure 7 IO Tab

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9. Next, Go to Ethernet Boot loader folder and launch “ EthernetBootloader_full imageloading.exe”

 Ethernet Bootloader	8/28/2014 12:08 PM	File folder	
 dotNetFx40_Client_x86_x64.exe	10/22/2013 3:17 PM	Application	41,993 KB
 PSP 2 software.7z	1/29/2016 2:20 PM	7Z File	126,914 KB
 StCamSWare_x64.exe	4/4/2014 8:38 AM	Application	11,318 KB
 StTrgDisplayVC2005.exe	2/4/2013 10:25 AM	Application	595 KB
 uEye64_38200_WHQL.exe	4/4/2014 8:37 AM	Application	79,672 KB



(E:) > WI_Latest > V810 PSP series 2 > PSP 2 software > Ethernet Bootloader				
	ified	Type	Size	
 fwfirmwarepatchforenableanddisablepro...	8/28/2014 12:08 PM	File folder		
 20130709-Fix Image Flickering ENC.afx	7/9/2013 11:47 AM	AFX File	205 KB	
 ENC_FullImageSupport.hex	6/6/2013 6:17 AM	HEX File	203 KB	
 ENC_TimeoutEnable.hex	5/29/2013 5:51 AM	HEX File	174 KB	
 EthernetBootloader.exe	6/10/2013 11:58 A...	Application	1,202 KB	
 EthernetBootloader.exe_backup	6/29/2013 5:12 AM	EXE_BACKUP File	1,272 KB	
 EthernetBootloader_fullImageLoading.exe	6/6/2013 2:28 AM	Application	1,305 KB	
 FirmwareUploader.exe	6/11/2013 2:58 AM	Application	1,202 KB	
 Latest_ENC.afx	6/20/2013 12:11 PM	AFX File	204 KB	

Figure 8 Ethernet Bootloader

10. If error prompt after launching this software where complaining “Missing. Net Framework”, please install “dotNetFx40_Full_x86_x64.exe “.

If able to open “EthernetBootloader_full imageloading.exe “, please direct go to Step 11.

PSP 2 software				
Name	Date modified	Type	Size	
 Ethernet Bootloader	8/28/2014 12:08 PM	File folder		
 dotNetFx40_Client_x86_x64.exe	10/22/2013 3:17 PM	Application	41,993 KB	
 PSP 2 software.7z	1/29/2016 2:20 PM	7Z File	126,914 KB	
 StCamSWare_x64.exe	4/4/2014 8:38 AM	Application	11,318 KB	
 StTrgDisplayVC2005.exe	2/4/2013 10:25 AM	Application	595 KB	
 uEye64_38200_WHQL.exe	4/4/2014 8:37 AM	Application	79,672 KB	

Figure 9 Dot Net Software

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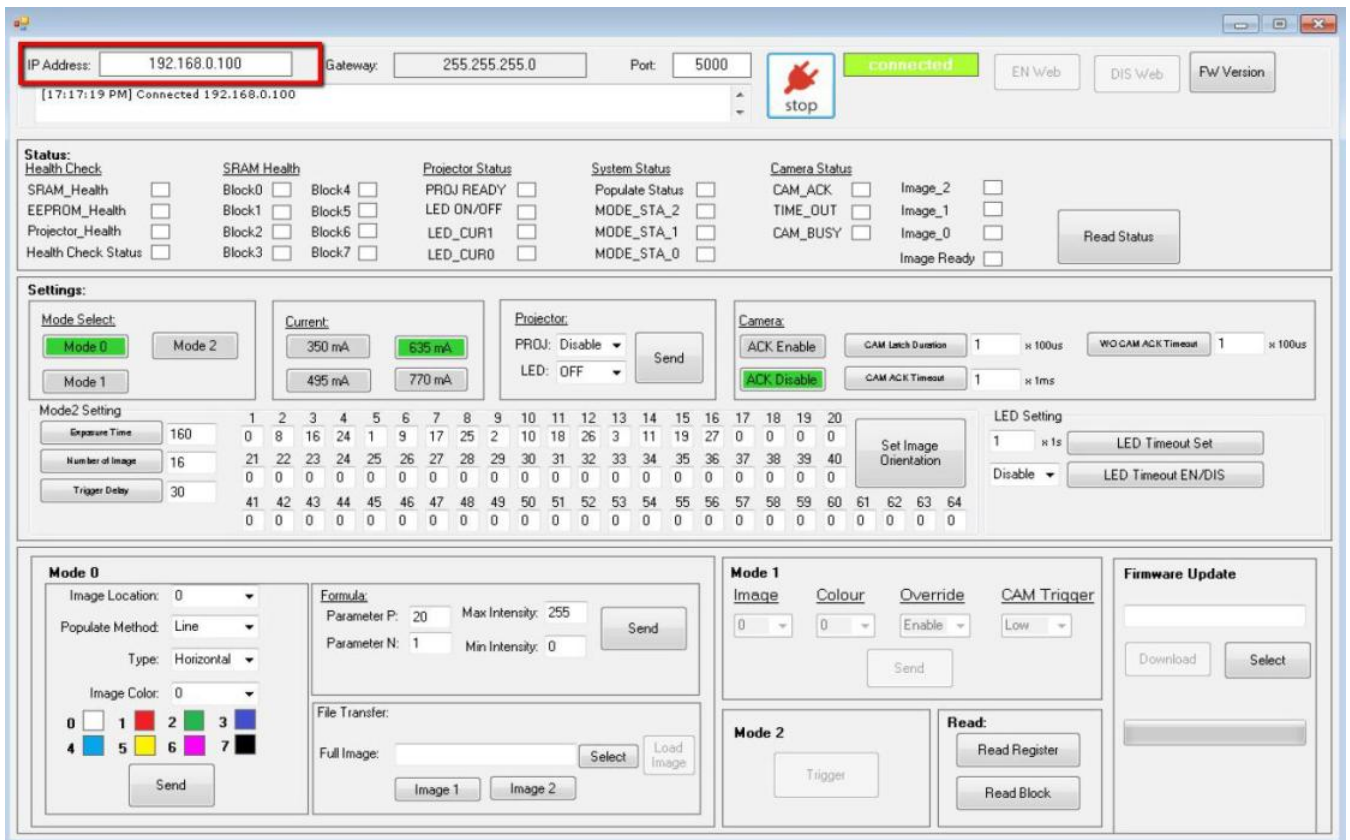
11. To Connect Front Module, please use the IP address from Step 1 to apply at the red bounding box which illustrated at below :

“Remarks: End IP address with. 100 means front module;

End IP address with .101 means rear module “

Type	IP ADDRESS (Front MODULE)	IP ADDRESS (REAR MODULE)
OLD MACHINE	192.168.0.100	192.168.0.101
NEW MACHINE	123.123.123.100	123.123.123.101

Click Connect.

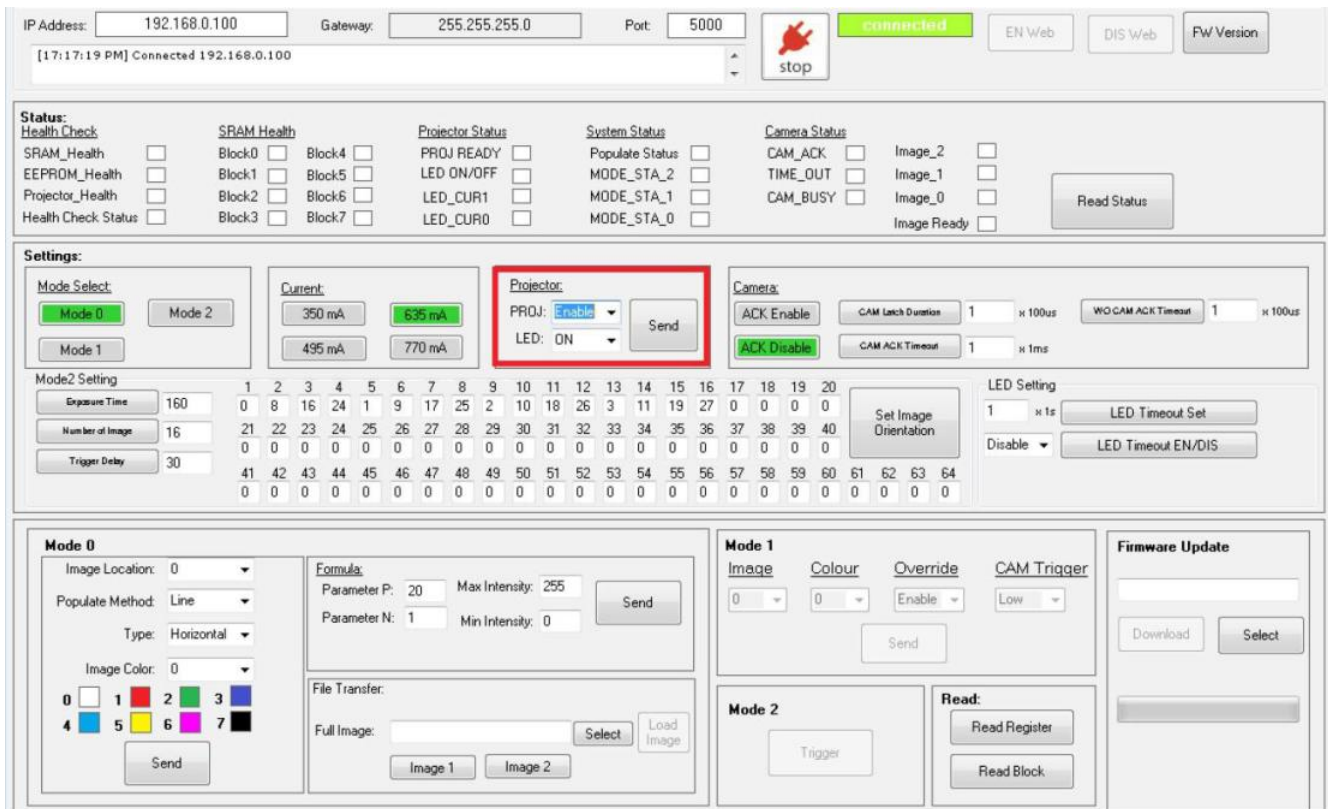


The screenshot shows the Ethernet Boot loader GUI. At the top, the IP Address field is highlighted with a red box and contains '192.168.0.100'. The Gateway is '255.255.255.0' and Port is '5000'. The status is 'connected'. The interface includes sections for Status (Health Check, SRAM Health, Projector Status, System Status, Camera Status), Settings (Mode Select, Current, Projector, Camera, Mode2 Setting, LED Setting), and Mode 0/1/2 configuration options.

Figure 10 Ethernet Boot loader GUI (IP ADDRESS Reference Only)

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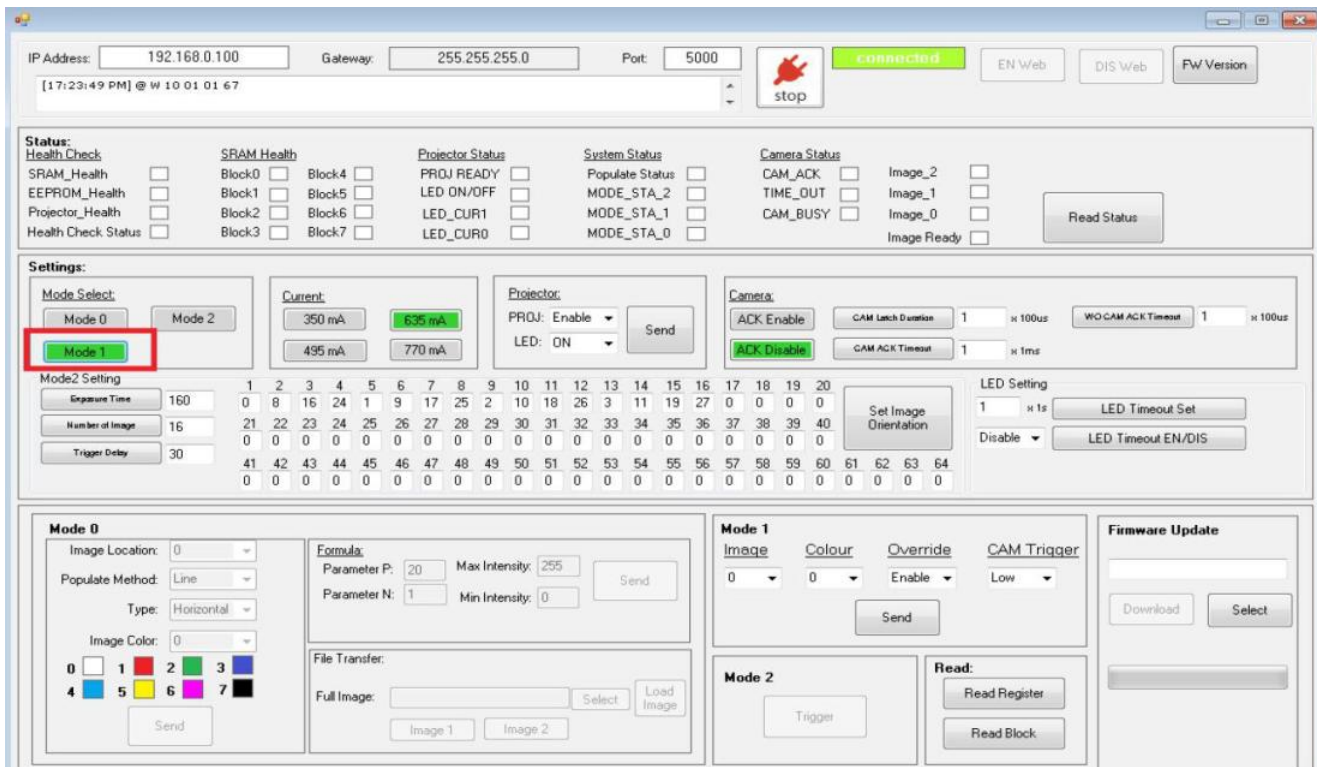
12. Then, Choose “ENABLE” at PROJ & Choose “ON” at LED & Press “Send” as below Figure 11.



The screenshot shows the Ethernet Bootloader GUI. At the top, there's a header with IP Address (192.168.0.100), Gateway (255.255.255.0), Port (5000), and a 'connected' status. Below this is a 'Status' section with various health checks. The 'Settings' section is the main focus, showing 'Mode Select' (Mode 0), 'Current' (635 mA), 'Projector' (set to 'Enable'), and 'LED' (set to 'ON'). A red box highlights the 'Projector' and 'LED' settings along with the 'Send' button. Other settings include 'Camera' (ACK Enable), 'Mode2 Setting' (Exposure Time 160, Number of Image 16, Trigger Delay 30), and 'LED Setting' (LED Timeout Set, LED Timeout EN/DIS). The bottom section shows 'Mode 0' and 'Mode 1' configurations, including image location, populate method, and file transfer options.

Figure 11 Ethernet Bootloader GUI

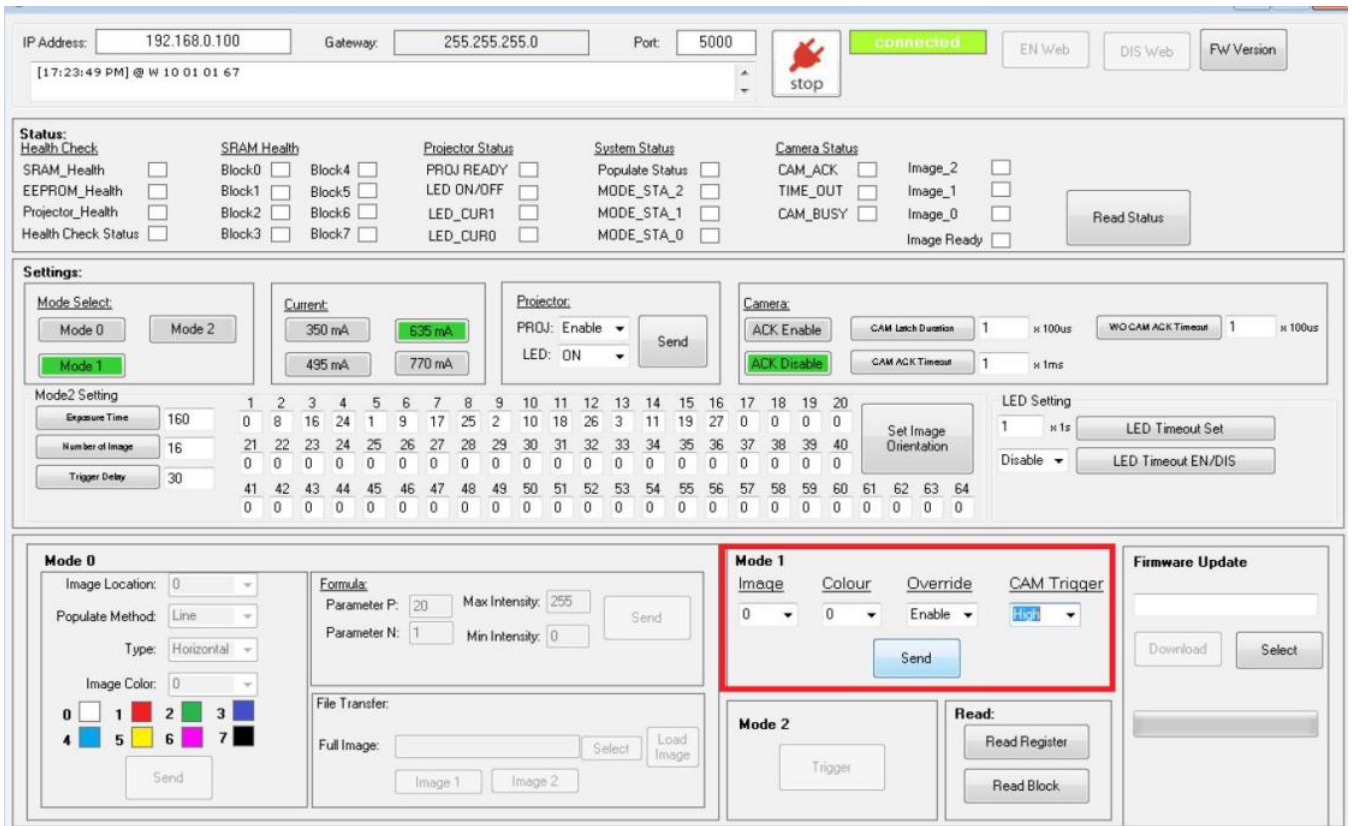
13. Then, Click “Mode 1 Button” as below Figure 12.



This screenshot is similar to Figure 11, but the 'Mode 1' button in the 'Mode Select' section is highlighted with a red box. The 'Projector' dropdown remains set to 'Enable' and the 'LED' dropdown remains set to 'ON'. The 'Send' button is still visible. The rest of the GUI, including the status section, mode settings, and file transfer options, remains the same as in Figure 11.

Figure 12 Ethernet Bootloader GUI

14. Next, under Mode 1 Section, select “High” for Cam Trigger & Choose “Enable” at Override as below Figure 13. Press Send after selection completed.



The screenshot shows the Ethernet Bootloader web interface. At the top, there's a header with IP Address (192.168.0.100), Gateway (255.255.255.0), Port (5000), and a 'connected' status. Below this is a 'Status' section with various health and system checks. The 'Settings' section includes Mode Select (Mode 0, Mode 2, Mode 1), Current (350 mA, 635 mA, 495 mA, 770 mA), Projector (PROJ: Enable, LED: ON), and Camera (ACK Enable, ACK Disable, CAM Latch Duration, CAM ACK Timeout). The 'Mode 1' section is highlighted with a red box, showing 'Image' (0), 'Colour' (0), 'Override' (Enable), and 'CAM Trigger' (High). The 'Send' button is visible. Other sections include Mode 2 Setting, File Transfer, and Firmware Update.

Figure 13 Ethernet Bootloader

15. If the Triggering cable is Functioning, screen shoot will be able to illustrate at “StTrgDisplayVC2005” “software as below Figure 14.

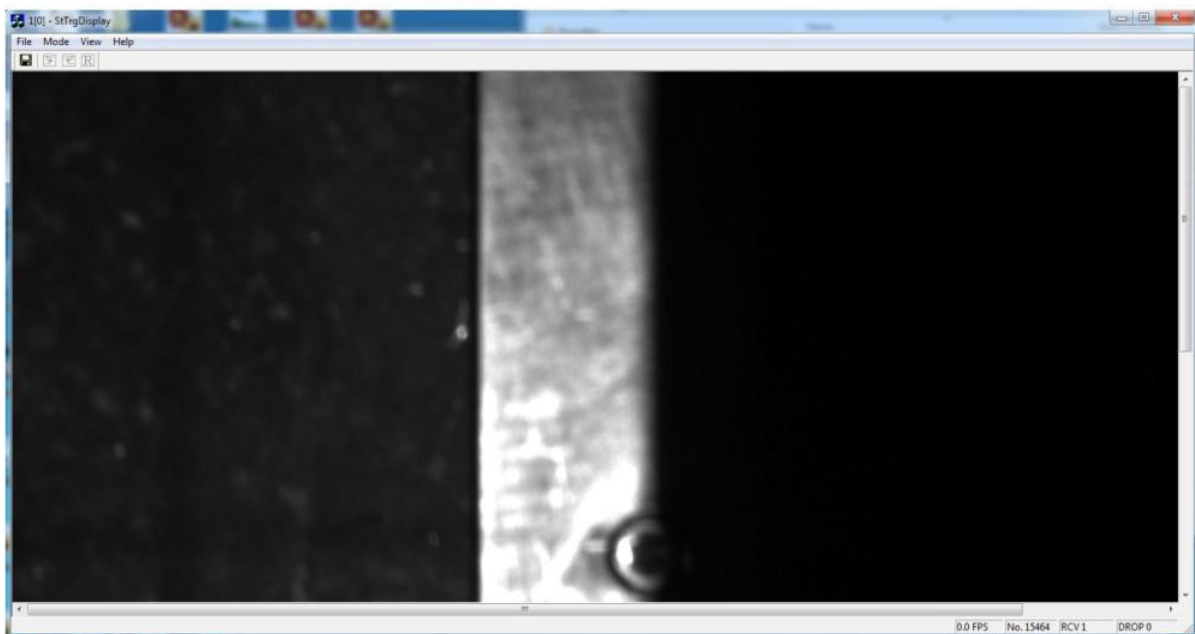


Figure 14 StTrgDisplayVC2005 GUI

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16. Once Step 14 completed, please revert the setting as Origin setting at “StTrgDisplayVC2005
“Software as below Figure 15. Go to Trigger Mode & Select “Free Run”.

The screenshot shows the 'Setting' window for 'StTrgDisplayVC2005'. The 'Trigger Mode' tab is active. The 'Trigger Type' dropdown menu is set to 'Free Run' and is highlighted with a red rectangle. Other settings include 'Trigger Source', 'Exposure Time', 'Auto Sub Trigger', 'Noise Reduction', 'Wait HD', 'Wait Read Out', 'Read Out Source', 'Exposure', 'Camera Memory', 'Trigger Mask', and 'CMOS Reset Type'. Buttons for 'Trigger', 'Sub Trigger', 'Read Out', 'Reset Frame No', 'OK', 'Save...', 'Load...', and 'Refresh' are visible.

Figure 15 StTrgDisplayVC2005 Setting

17. Next, Go to IO tab at row IO0, under mode, select “Disable” as below Figure 16.

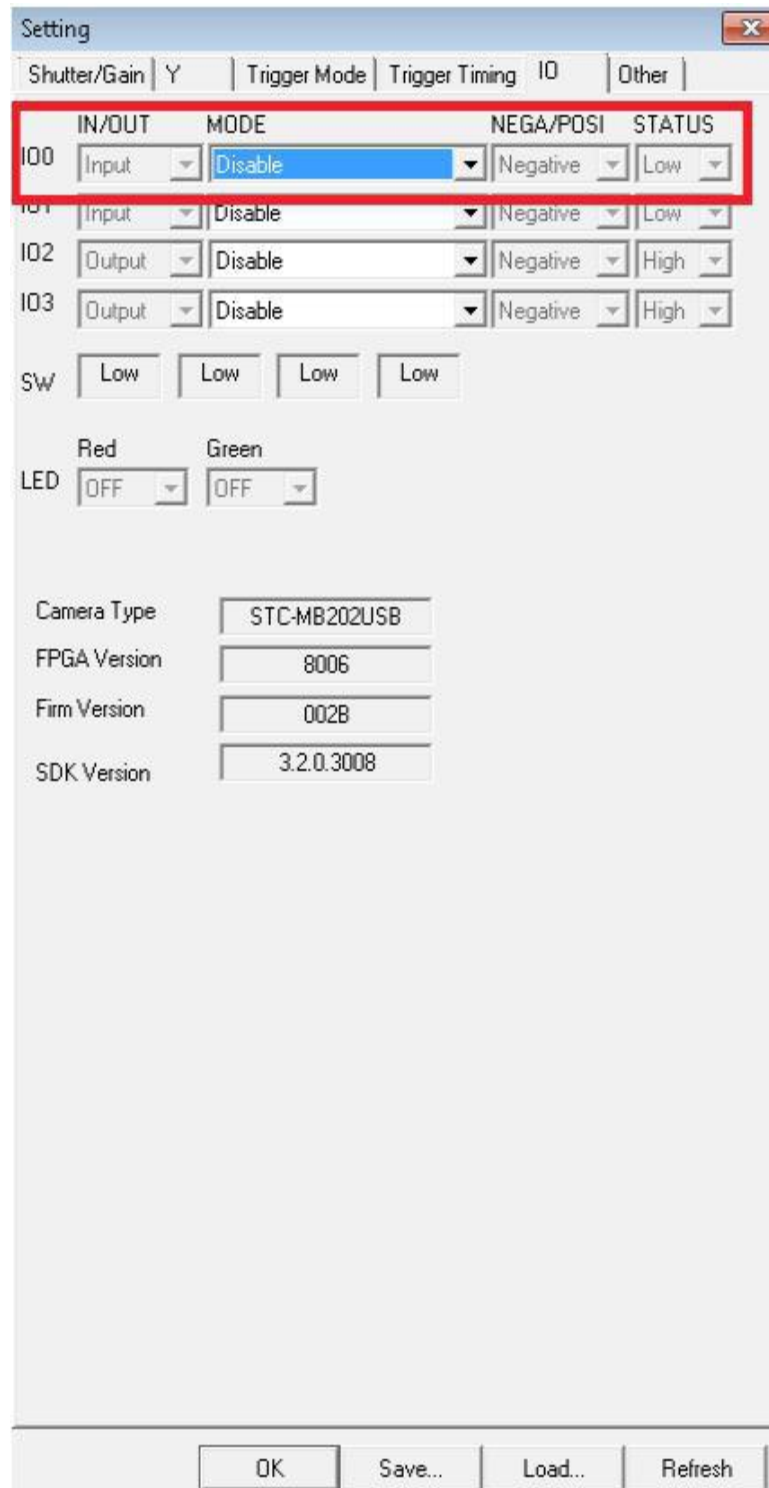


Figure 16 StTrgDisplayVC2005 Setting

18. Repeat the Step 1 ~ 16 for Rear module Checking by using “IP address which end with .101” at HW Config.